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ORIGINAL RESEARCH REPORT



Ultrasound stress compromises the correlates of emotional-like states and brain AMPAR expression in mice: effects of antioxidant and anti-inflammatory herbal treatment

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ABSTRACT

The modern lifestyle is associated with exposure to “psychological” or “emotional” stress. A growing portion of the population is exposed to emotional stress that results in a high incidence of anxiety disorders, a serious social problem. With this rise, there is a need for understanding the neurobiological causes of stress-induced anxiety and to offer safe remedies for this condition. Side effects of existing pharmaceuticals necessitate the search for alternatives. Having fewer adverse effects than classic remedies, natural extract-based therapies can be a promising solution. Here, we applied a model of emotional stress in BALB/c mice using ultrasound exposure to evoke the signs of anxiety-like behavior. We examined the behavioral and molecular impact of ultrasound and administration of herbal antioxidant/anti-inflammatory treatment (HAT) on AMPA receptor expression, markers of plasticity, inflammation and oxidative stress. A 3-week ultrasound exposure increased scores of anxiety-like behaviors in the standard tests and altered hippocampal expression as well as internalization of AMPA receptor subunits GluA1–A3. Concomitant treatment with HAT has prevented increases of anxiety-like behaviors and other behavioral changes, normalized hippocampal malondialdehyde content, GSK3 β and pro-inflammatory cytokines IL-1 β and IL-6, and the number of Ki67-positive cells. Levels of malondialdehyde, a common measure of oxidative stress, significantly correlated with the investigated end-points in stressed, but not in non-stressed animals. Our results emphasize the role of oxidative stress in neurobiological abnormalities associated with experimentally induced condition mimicking emotional stress in rodents and highlight the potential therapeutic use of anti-oxidants like herbal compositions for management of stress-related emotional disturbances within the community.

ARTICLE HISTORY

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KEYWORDS

Ultrasound stress; anxiety-like behavior; oxidative stress; AMPA glutamate receptors; pro-inflammatory cytokines; mice

1. Introduction

The incidence of anxiety disorders is currently the highest of all mental disorders (Kessler, Petukhova, Sampson, Zaslavsky, & Wittchen, 2012; Bandelow & Michaelis, 2015). As for instance, 20% of adults are diagnosed with generalized anxiety alone (Grenier et al., 2019). Anxiety disorders are frequently accompanied by increased aggression, violence, self-harm, and suicide (Stone, 2016). “Psychological” or “emotional” stress resulting from the adverse experience of non-material nature is the most common cause of these conditions (Fontes et al., 2014). At the same time, side effects of traditional anti-anxiety drugs are well-documented: benzodiazepines, the most widely used

anxiolytics, cause rebound anxiety, insomnia and memory impairments (Uzun, Kozumplik, Jakovljević, & Sedić, 2010); barbiturates induce drowsiness and balance problems, disrupt learning and reproductive functions (Anderson & Hakimian, 2018), selective serotonin re-uptake inhibitors (SSRIs) evoke irritability and restlessness (Orsolini & Bellantuono, 2015), as well as fatigue, nausea and insomnia (Quagliato et al., 2019). That said, there is a need for an effective and safe pharmacological treatment, and herbal medicine appears as an attractive solution (Bystritsky et al., 2012) affordable for low-income patients (Oyebode, Kandala, Chilton, & Lilford, 2016). However, with an increase in self-medication with herbal remedies, there is a

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need for better understanding of their mechanisms of action to prevent potential adverse effects (McIntyre, Saliba, & Moran, 2015).

Herbal medicine is known to normalize behavioral correlates of anxiety-like state and stress-associated changes experimentally induced in small rodents. This was shown in mice receiving chlorogenic acid extract from *Prunus domestica* (Bouayed, Rammal, Dicko, Younos, & Soulimani, 2007) or *Beta vulgaris* during restraint stress (Sulakhiya et al., 2016) and in rats treated with an extract from *Ulva spiralis* (Violle, Rozan, Demais, Nyvall Collen, & Bisson, 2018). Herbal treatments with stress-reducing properties diminish oxidative stress, one of the causes of anxiety (Salim, 2017; Umukoro, Adebisin, Agu, Omorogbe, & Asehinde, 2018) and aggression (Schiavone, Colaianna, & Curtis, 2015).

Here, we investigated the potential anti-stress/anti-anxiety properties of a herbal antioxidant/anti-inflammatory treatment (HAT), a composition, containing seaweed, ginger, lemon, orange elderberries and other elements at the concentrations exploited in human and animal studies (Supplementary Table 1). A recently established rodent model of “emotional stress” was used, where mice and rats are exposed to the ultrasound of alternating frequencies (25–45 kHz) that mimics rodent vocalizations emitted during conditions associated with “negative emotional state” (Gorlova et al., 2019; Morozova et al., 2016; Pavlov et al., 2019). A 3-week exposure of BALB/c mice to ultrasound stress increases behavioral correlates of aggressive behavior, anhedonia and helplessness. Ultrasound stress also increases levels of pro-inflammatory cytokines and Glycogen-Synthase-kinase-3-beta (GSK3 β) in the brain, oxidative stress markers glutathione and malondialdehyde, decreases the levels of plasticity markers, and dysregulates gene expression of AMPA receptor subunits GluA1–A3, but not GluA4 (Gorlova et al., 2019; Pavlov et al., 2019). Compounds with anti-oxidative properties prevent most of these changes (Gorlova et al., 2019).

In this study, we investigated the effect of ultrasound stress on the behavioral signs of anxiety-like behavior and internalization of GluA1–A3 AMPA receptor subunits in the hippocampus, one of the key brain structures involved in emotional regulation during stress (Campbell & MacQueen, 2004), in mice. We additionally investigated whether above-mentioned new herbal antioxidant/anti-inflammatory treatment (HAT) can affect these measures induced by ultrasound stress, and alter hippocampal content of oxidative stress marker malondialdehyde (Ayala, Muñoz, & Argüelles, 2014), as it was suggested by previous studies with applied here model (Gorlova et al., 2019; Pavlov et al., 2019). These changes were related with the above-listed end-points and to hippocampal expression of GSK-3 β , and Akt, two reciprocally related regulators of distress, plasticity and neuroinflammation (Beurel, Grieco, & Joje, 2015), as well as cytokines: interleukin (IL) IL-1 β , IL-6, IL-15, tumor necrosis factor (TNF), and to the aggression scores.

2. Materials and methods

2.1. Animals

BALB/c male mice, 10–12-week-old, were obtained from certified Charles River provider and housed individually under

standard laboratory conditions ($22 \pm 1^\circ\text{C}$, 55% humidity, reversed 12-h light/dark cycle with lights on at 19:00, food and water *ad libitum*). Experimental procedures were set up in accordance with Directive 2010/63/EU and approved by the veterinarian committee of C.Bernard University. All efforts were undertaken to minimize the potential discomfort of the animals.

2.2. Study design

Mice were weighed and assigned to one of six experimental groups: (1) control (non-stressed) non-treated mice, (2) control mice treated with HAT-solvent, (3) control HAT-treated mice, (4) stressed non-treated mice, (5) stressed mice treated with HAT-solvent, (6) stressed HAT-treated mice, each comprising in six to eight animals. Mice were subjected to a 3-week ultrasound of unpredictably alternating frequencies 24/7 (see below), while receiving respective treatments, body weight was measured. Twenty-four hours after, their behavior was subsequently investigated in the O-maze, open field and resident-intruder paradigms, with 2-h inter-test intervals. Applied experimental scheme was designed to reduce the potential stress of the behavioral testing. Mice were killed 16 h after the end of the last behavioral test. Brains and right and left hippocampi were dissected using previously established method (Cline et al., 2012; Pavlov et al., 2019; see below) for immunohistochemical study of AMPA receptor subunits GluA1–A3, neuronal marker NeuN and proliferation marker Ki67, or quantitative reverse transcriptase PCR (qRT-PCR) of IL-1 β , IL-6, TNF, GSK3 β , Akt and malondialdehyde (MDA) assays, respectively (Figure 1; see below). Above-described study design was adapted from similar experimental settings, which previously showed lack of effects of the behavioral testing alone on the measured parameters of stress, plasticity and neuroinflammation in mice (Cline et al., 2015; Couch et al., 2013, 2016; Costa-Nunes et al., 2014; Gorlova et al., 2019; Vignisse et al., 2017).

2.3. Treatment administration

Solutions of the herbal treatment (HAT) consisting of S-A and S-D drops, or its HAT-solvent (provided by Vendeville, Voorhout, The Netherlands; Supplementary Table 1) were administered orally using a pipette for 21 days concomitantly with ultrasound emission. Herbal treatment was dosed twice: administration of 30 μL of S-A solution was followed by the administration of 90 μL of S-D solution with a half-hour interval. In total, each HAT-treated mouse received 120 μL of herbal treatment. Groups of mice treated with HAT-solvent, alcohol gluten-free based solution (Supplementary Table 1), received 120 μL of this solution. Concentrations of HAT ingredients were in a range of herbal treatments used alone or in herbal compositions used in human and in animal studies (see Supplementary Table 2).

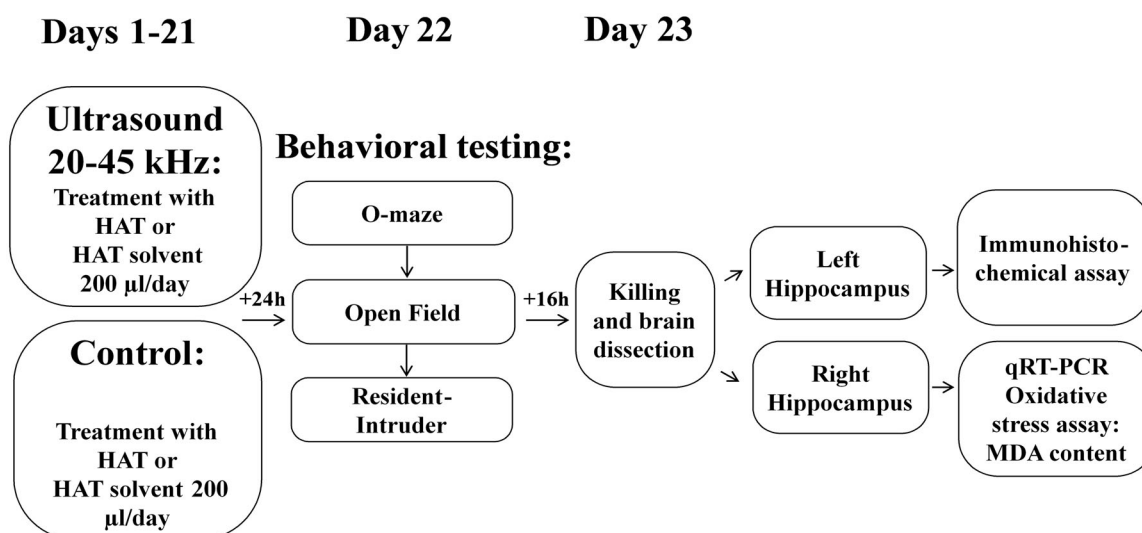


Figure 1. Experiment design. After 21-day of ultrasound exposure and dosing with herbal treatment (HAT) or HAT-solvent, mice were tested for behavior and then killed; their hippocampi were dissected for *in vitro* assays, including quantitative reverse transcription polymerase chain reaction (qRT-PCR) a measurement of malondialdehyde (MDA) content.

2.4. Ultrasound emission

Ultrasound emission of average intensity of 50 ± 5 dB and frequencies in a 20–45 Hz range was continuously delivered to the experimental groups of mice using a random schedule of alternating frequencies via a manufactured device (Weitech, Wavre, Belgium) and controlled by a detector (Discovery Channel, Crozet, VA, USA) as described elsewhere (Gorlova et al., 2019; Pavlov et al., 2019, see [Supplementary material, Supplementary Figure 1](#)). During the 10-min periods, ultrasound frequencies fluctuated at short time spans of ≤ 1 s. 35% of the emission time consisted of 20–25 and 25–40 kHz frequencies, and 30% of the emission time consisted of 40–45 kHz frequencies. The ultrasound device was suspended 2 m above the cages of the experimental groups with an average horizontal distance to cages of 2.5 m. The position of the cages with respect to the stimulator was changed weekly. Applied ultrasound emission induced a loss of body weight and an increase of blood corticosterone concentrations in the experimental mice ([Supplementary Figure 3](#)) that demonstrates its stress impact. Previous validation studies suggested the selectivity of the adverse effects of applied low-frequency ultrasound versus the potential general negative action of a constant noise on behavioral correlates of “emotional state” in mice (Gorlova and Pavlov, *personal communication*). In particular, when mixed frequencies (“white noise”) at the range of 20 Hz–20 kHz were employed for 3 weeks, no changes in anxiety-like behaviors in the elevated maze and open field were observed (Gorlova et al., *personal communication*).

2.5. O-maze test

The apparatus (Open Science, Russia), which consisted of a circular path (runway width 5.5 cm, diameter 46 cm), was placed 50 cm above the floor. Two opposing arms were protected by walls (height 10 cm), and the illumination strength was 25 lx. Anxiety-like behavior was assessed using previously

validated parameters (Cline et al., 2015; Couch et al., 2016; Strekalova et al., 2016). Mice were placed in one of the closed arm compartments of the maze. The number of events of risk assessments, the number of exits to the open arms of the maze, total duration of time spent therein and the number of crossed sectors (4×5.5 cm) were scored during 5 min.

2.6. Open field test

In the open field model, mice were placed in a square gray plastic box ($45 \times 45 \times 45$ cm) illuminated with white light (25 lx) near the wall. The number of exploratory rearings, the number of grooming events, time spent in the central zone (15×15 cm) and the number of crossed sectors (11.25×11.25 cm) were scored for 10 min as described elsewhere (Cline et al., 2012; Pavlov et al., 2019; Strekalova et al., 2016). The recordings were subsequently analyzed using the automated video-tracking Viewpoint software (Viewpoint, Lyon, France).

2.7. Resident-intruder test

In the resident-intruder test, mice were individually placed in an observation cage ($30 \times 60 \times 30$ cm) for 30 min. Then, each mouse was exposed to a previously group-housed naïve CD1 male intruder of similar weight and age for 4 min in the same cage. The latency to attack and their total number of attacks were scored as described elsewhere, number of attacking animals per group was calculated (Costa-Nunes et al., 2014; Gorlova et al., 2019; Strekalova et al., 2018).

2.8. Brain collection

Mice were terminally anesthetized with an intraperitoneal injection of Nembutal (Bayer, Wiesbaden, Germany) and then transcardially perfused with 10 mL of the ice-cold 0.9% NaCl. Thereafter, right hippocampi of each mice were isolated via

biopsy and immediately frozen at -80°C on dry ice. The biopsy was performed by opening a section of the skull and cutting the hippocampus from above without damaging the basal arteries. After that, mice were transcardially perfused by 4%-paraformaldehyde (PFA) in 0.1 M phosphate buffer. Left hippocampi were removed and postfixed in 4%-PFA for 12 h at $+4^{\circ}\text{C}$, immersed in 0.1 M phosphate-buffered saline (PBS, pH 7.4) for 12 h at $+4^{\circ}\text{C}$, and processed for paraffin embedding; all procedures were carried out as described elsewhere (Cline et al., 2015; Pavlov et al., 2019). 10 mg of frozen tissue was separated for MDA assay, and the rest of the tissue was used for RT-PCR analysis.

2.9. Quantitative reverse transcription polymerase chain reaction analysis (qRT-PCR)

Quantitative real-time PCR (qRT-PCR) was carried out as described elsewhere (Cline et al., 2012, 2015; Couch et al., 2016; Pavlov et al., 2019). Briefly, total mRNA was isolated with TRI Reagent (Invitrogen, Carlsbad, CA, USA). 1 μg of total RNA was converted into cDNA using random primers and Superscript III transcriptase (Invitrogen, Carlsbad, CA, USA). qRT-PCR was performed in a 10 μL reaction volume containing a SYBR Green master mix (5 μL , Bio-Rad Laboratories, Philadelphia, PA, USA). All qRT-PCR samples were ran on a CFX96 Deep Well Real-Time PCR Detection System (Bio-Rad, Hercules, CA, USA) in triplicate (see [Supplementary material](#) and [Supplementary Table 3](#)). The housekeeping gene glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was selected as the reference following previously established protocols (Cline et al., 2015; Pavlov et al., 2019). Data were normalized to GAPDH mRNA expression and calculated as relative-fold changes compared to control mice as described elsewhere (Couch et al., 2016; Gorlova et al., 2019; Strekalova et al., 2016).

2.10. Immunohistochemical analysis

Immunohistochemical analysis was performed as described previously (Pavlov et al., 2019; Strekalova et al., 2018; Vignisse et al., 2017). Ki67 immunofluorescent staining was performed to identify proliferating cells; hexaribonucleotide binding protein-3 (NeuN) immunofluorescent staining was used to label neurons, staining with DAPI was performed to visualize cell nuclei. Immunofluorescent staining of subunits of AMPA receptor was performed for subunits GluA1-GluA3. Numbers of surface puncta and of internalized puncta, and the internalization index were determined for each subunit. An internalization index was calculated by dividing the number of internalized puncta by the sum of internalized and surface numbers of puncta (see [Supplementary material](#)).

2.11. Malondialdehyde assay

Hippocampal malondialdehyde content was measured following Abcam ab118970 kit instructions (Abcam, Eugene, OR, USA; see [Supplementary material](#)). Briefly, the tissue was washed in cold PBS and homogenized in lysis solution, and then centrifuged at 13,000 g for 10 min. TBA reagent was

added to supernatant and, after incubation at 95°C for 60 min and cooling to RT in an ice bath, supernatant was analyzed at 532 nm in a 96-well microplate as described elsewhere (Pavlov et al., 2019; Vignisse et al., 2017).

2.12. Statistical analysis

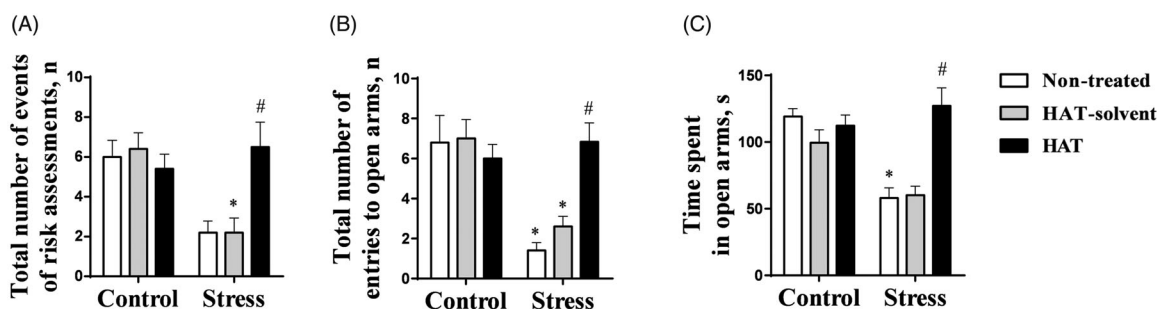
Data analysis was performed using GraphPad Prism software version 6.01 for Windows (GraphPad Software, La Jolla, CA, USA). Data were treated by a two-way ANOVA with Tukey's multiple comparison post-hoc test. Cohen's d for each significant finding to provide effect sizes was additionally calculated. The χ^2 -test was applied to compare the number of attacking vs non-attacking animals between the groups. Spearman test was used for correlation analysis. Statistical significance was set at $p < 0.05$. Data are shown as mean $\pm$ SEM.

3. Results

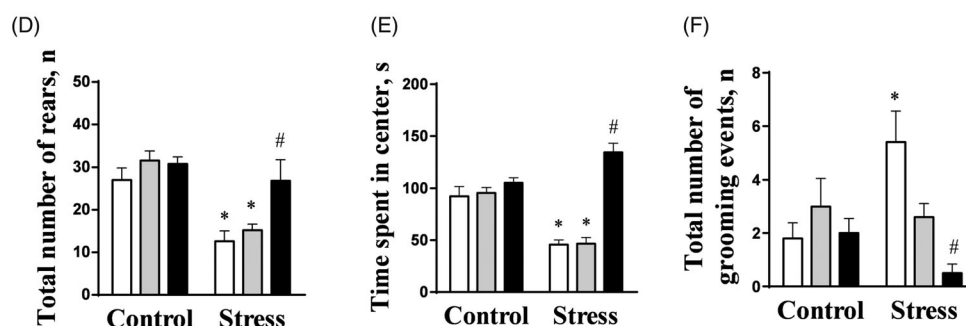
3.1. Effects of ultrasound exposure and herbal treatment in the O-maze

ANOVA revealed a significant interaction between ultrasound exposure and herbal treatment for total number of events of risk assessments, total number of entries to the open arms of the O-maze, and total time spent by mice in the open arms ($F_{2,25} = 5.568$, $p = 0.0100$; $F_{2,25} = 7.405$, $p = 0.003$; $F_{2,25} = 8.852$, $p = 0.001$, respectively, two-way ANOVA). All these parameters were significantly affected by stress ($F_{1,25} = 9.857$, $p = 0.004$; $F_{1,25} = 17.22$, $p < 0.001$; $F_{1,25} = 13.67$, $p = 0.001$, respectively) and by herbal treatment ($F_{2,25} = 6.635$, $p = 0.0091$; $F_{2,25} = 3.718$, $p = 0.039$; $F_{2,25} = 10.20$, $p < 0.001$, respectively, two-way ANOVA; [Figure 2\(A–C\)](#)). Post hoc analysis revealed a significant decrease in the total number of entries to the open arms, the time spent in the open arms of the O-maze, and non-significant decrease in the total number of events of risk assessments in stressed non-treated mice in comparison with control non-treated mice ($p = 0.003$ and $d = 3.71$; $p = 0.002$ and $d = 3.98$; $p = 0.066$, Tukey's and Cohen's d tests, respectively). There was a significant decrease in the total number of events of risk assessments, the total number of entries to open arms, and a non-significant decrease in the time spent in the open arms of the O-maze in stressed mice treated with HAT-solvent in comparison with control non-treated mice ($p = 0.033$, and $d = 2.69$; $p = 0.021$ and $d = 2.96$; $p = 0.073$, respectively) that suggested a lack of significant effects of the solvent on measured behavioral parameters of stressed mice. These parameters were significantly higher in stressed HAT-treated mice in comparison with stressed non-treated animals ($p = 0.019$ and $d = 1.96$; $p = 0.002$ and $d = 3.38$; $p < 0.001$ and $d = 2.78$; [Figure 2\(A–C\)](#)). These data suggest increases of behavioral correlates of pro-anxiogenic changes in the ultrasound-exposed mice, their rescue by herbal treatment, and the lack of significant effects of its solvent on this behavior. No differences were found between groups of control non-stressed mice, suggesting a lack of effects of the solvent and HAT in these animals.

O-MAZE



OPEN FIELD



RESIDENT-INTRUDER

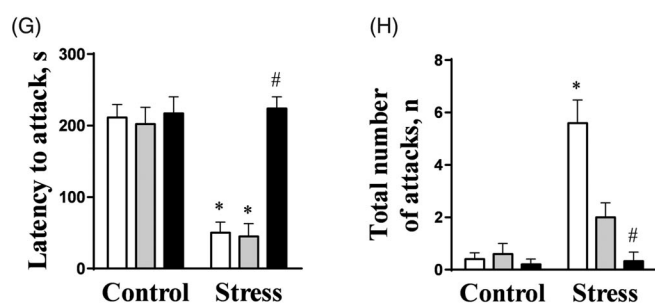


Figure 2. Behavioral changes in the ultrasound-exposed mice and effects of herbal treatment. In the O-maze test, in comparison to control non-treated group, stressed non-treated group and mice treated with HAT-solvent showed a significant decrease in (A) the total number of events of risk assessments (B) total number of entries to open arms and (C) time spent in open arms. Stressed HAT-treated animals did not differ from control non-treated mice in these measures. In the open field test, in comparison with control non-treated mice, stressed non-treated mice and stressed mice treated with HAT-solvent showed significant decreases in (D) total number of rears and (E) time spent in the center and (F) a significant increase in the number of grooming events. Stressed HAT-treated animals showed no such differences. In the resident-intruder test, in comparison with control non-treated group, non-treated stressed mice and stressed mice treated with HAT-solvent showed (G) a significant decrease of latency to attack and (H) a significant increase in the total number of attacks. Again, no such changes were found in stressed animals that received herbal treatment. * $p < 0.05$ vs. non-treated control group, # $p < 0.05$ vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used. HAT: herbal antioxidant treatment. Bars are Mean $\pm$ SEM.

We found no significant effects of stress ($F_{1,24} = 0.113$, $p = 0.739$, two-way ANOVA), treatment ($F_{2,24} = 0.147$, $p = 0.863$), or interaction ($F_{2,24} = 1.004$, $p = 0.381$; [Supplementary Figure 4\(A\)](#)) on the number of crossed sections in the O-maze.

3.2. Altered open field behavior in mice subjected to the ultrasound and effects of herbal treatment

In the open field test, there was a significant interaction between ultrasound exposure and herbal treatment on the total number of rearings and the time spent in the center ($F_{2,25} = 0.247$, $p = 0.105$ and $F_{2,25} = 20.58$, $p < 0.001$,

respectively, two-way ANOVA), as well as significant stress and treatment effects ($F_{1,25} = 21.71$, $p < 0.001$ and $F_{2,25} = 4.531$, $p = 0.0209$, $F_{1,25} = 15.03$, $p < 0.001$ and $F_{2,25} = 34.67$, $p < 0.001$, respectively). Post hoc analysis showed significant decrease in the number of rearings and the total time spent in center in stressed non-treated mice ($p = 0.031$ and $d = 2.45$; $p = 0.001$ and $d = 4.01$, respectively; Tukey's and Cohen's d tests) and mice treated with HAT-solvent ($p = 0.011$ and $d = 2.50$; $p < 0.001$ and $d = 3.70$, respectively) in comparison to control non-treated mice. This measures were increased in stressed HAT-treated mice in comparison with stressed non-treated animals ($p = 0.024$ and $d = 1.53$; $p < 0.001$ and $d = 5.49$; [Figure 2\(D,E\)](#)).

There was no significant interaction between effects of the ultrasound exposure and treatment on this parameter ($F_{2,25}=0.8759$, $p=0.3583$, two-way ANOVA), although we found a significant effect of herbal treatment ($F_{2,25}=5.311$, $p=0.012$) and stress ($F_{2,25}=6.534$, $p=0.005$) on the number of grooming events. Stressed HAT-treated animals displayed a decrease in the number of grooming events in comparison with stressed non-treated group ($p<0.001$ and $d=2.84$; Tukey's and Cohen's d tests; Figure 2(F)). Of note, solvent-treated stressed animals showed no increase in this behavior suggesting that solvent, which was an alcohol-based solution, might have a partial anti-stress effect.

We found no significant effect of stress ($F_{1,24}=1.11$, $p=0.302$, two-way ANOVA), as well as no significant effect of treatment ($F_{2,24}=1.064$, $p=0.360$), or interaction ($F_{2,24}=0.0062$, $p=0.939$; Supplementary Figure 4(B)) on the number of crossed sections in the open field, that suggests unaltered locomotor activity in stressed animals. This findings rules out potential confounds in evaluation of anxiety like behaviors that can be associated with chronic stress exposure (Strekalova, Spanagel, Dolgov, & Bartsch, 2005).

Together, these results suggest that ultrasound stress increases scores of behavioral correlates of anxiety-like behavior, suppresses exploratory activity, induces excessive grooming, a sign of stress response, and that herbal treatment but not its solvent counteract these changes.

3.3. Effects of the ultrasound stress and herbal treatment on aggressive behavior

In the resident-intruder test, ANOVA revealed a significant interaction between ultrasound exposure and herbal treatment in latency to attack and the total number of attacks ($F_{2,25}=13.00$, $p<0.001$; $F_{1,25}=32.67$, $p<0.001$, respectively). These parameters were significantly affected by stress ($F_{1,25}=44.46$, $p<0.001$; $F_{1,25}=32.67$, $p<0.001$, respectively, two-way ANOVA) and herbal treatment ($F_{2,25}=16.45$, $p<0.001$; $F_{2,25}=16.60$, $p<0.001$, respectively). There was a significant decrease in latency to attack in non-treated stressed mice ($p<0.001$ and $d=4.37$, Tukey's and Cohen's d tests) and in stressed mice treated with HAT-solvent in comparison with control non-treated mice ($p<0.001$ and $d=4.13$). Stressed HAT-treated animals displayed decreased latency as compared to stressed non-treated group ($p<0.001$ and $d=4.75$; Figure 2(G)). Non-treated stressed mice also displayed a significant increase in the number of attacks in comparison with control non-treated mice ($p<0.001$ and $d=4.16$), while no such changes were observed for stressed HAT-treated animals ($p=0.147$) that also showed significantly smaller values of this measure, compared to the former group ($p<0.001$ and $d=3.81$, Figure 2(H)).

The χ^2 test showed a significant increase in number of attacking animals in non-treated stressed group and in stressed group treated with HAT-solvent in comparison with non-treated control group ($T=4.286$, $p=0.0384$ and $T=4.286$, $p=0.0384$, respectively). This measure was significantly decreased in the HAT-treated stressed group in comparison with non-treated stressed group ($T=7.639$,

$p=0.0057$, χ^2 test). Interestingly, solvent-treated stressed animals showed no significant increase in the number of attacks suggesting that solvent had partially prevented stress-induced augmentation of aggressive behavior. Together, behavioral correlates of aggressive behavior were increased in mice exposed to ultrasound that did not receive any treatment, but not in the stressed HAT-treated animals.

3.4. Ultrasound exposure changes expression and internalization of subunits of the AMPA receptor in the hippocampus: effects of herbal treatment

ANOVA revealed no significant interaction between the effects of stress and herbal treatment on mRNA expression of GluA1-A3 subunits ($F_{2,25}=1.298$, $p=0.230$; $F_{2,25}=2.301$, $p=0.121$; $F_{2,25}=0.2$, $p=0.820$). At the same time, there was a significant effect of stress on mRNA expression of GluA1-A3 subunits ($F_{1,25}=11.85$, $p=0.002$; $F_{1,25}=9.167$, $p=0.006$; $F_{1,25}=18.64$, $p<0.001$, respectively, two-way ANOVA) and no significant effect of herbal treatment ($F_{2,25}=3.11$, $p=0.062$; $F_{2,25}=2.93$, $p=0.072$; $F_{2,25}=2.048$, $p=0.150$, respectively). Non-significant decrease in the expression of GluA1 and GluA3 subunits was found in non-treated stressed mice ($p=0.102$; $p=0.099$, Tukey's test) and in stressed mice treated with HAT-solvent ($p=0.180$; $p=0.119$; Table 1) in comparison with control non-treated mice. There was a significant increase of GluA2 expression in stressed non-treated mice in comparison with control non-treated mice ($p=0.038$, and $d=1.93$, Tukey's and Cohen's d tests), but not in mice treated with HAT ($p=0.447$) or with HAT-solvent ($p=0.102$).

ANOVA revealed a significant stress and treatment interaction on number of internalized GluA1 and GluA2, but not GluA3 puncta ($F_{2,25}=9.115$, $p=0.001$; $F_{2,25}=4.264$, $p=0.0260$; $F_{2,25}=0.807$, $p=0.458$, respectively). There was a significant effect of stress on number of internalized GluA1, GluA2 and GluA3 puncta ($F_{1,25}=22.70$, $p<0.001$; $F_{1,25}=29.94$, $p<0.001$; $F_{1,25}=13.54$, $p=0.001$, respectively, two-way ANOVA), as well as significant effect of herbal

Table 1. Relative gene expression of GluA1, GluA2 and GluA3 subunits.

Groups	Relative-fold gene expression, % from control		
	GluA1	GluA2	GluA3
Control, non-treated (Con NT)	100 ± 4.24	100 ± 5.57	100 ± 3.68
Control, HAT-solvent (Con HATs)	104.7 ± 6.17	101.1 ± 9.08	111.6 ± 6.94
Control, HAT (Con HAT)	107.9 ± 10.67	98.1 ± 8.87	111.9 ± 7.13
US, non-treated (US NT)	67.2 ± 7.12	135.6 ± 10.95 vs Con NT * $p=0.038$	65.9 ± 10.85
US, HAT-solvent (US HATs)	75.4 ± 13.81	120.5 ± 6.08	78.8 ± 7.13
US, HAT (US HAT)	99.6 ± 5.37	100.7 ± 4.97	87.7 ± 8.91

Relative fold levels of mRNA expression of GluA1, GluA2 and GluA3 subunits normalized to control. There was a significant increase of GluA2 expression in stressed non-treated mice in comparison with control non-treated mice (* $p<0.05$ vs. non-treated controls, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used, relative mRNA expression data are expressed in percent from non-treated controls). US: ultrasound; HAT: herbal antioxidant treatment. Data are Mean ± SEM.

Table 2. Numbers of internalized GluA1, GluA2 and GluA3 puncta.

Groups	Internalized puncta, % from to control		
	GluA1	GluA2	GluA3
Control, non-treated (Con NT)	100 ± 20.49	100 ± 16.56	100 ± 18.01
Control, HAT-solvent (Con HATs)	94.0 ± 11.07	101.2 ± 14.41	103 ± 5.99
Control, HAT (Con HAT)	113.6 ± 17.28	89.4 ± 10.01	90 ± 16.02
US, non-treated (US NT)	227.8 ± 18.40 vs Con NT * <i>p</i> < 0.001	203.6 ± 22.54 vs Con NT * <i>p</i> = 0.003	163.8 ± 15.79 vs Con NT * <i>p</i> = 0.043
US, HAT-solvent (US HATs)	162.6 ± 12.33	164.6 ± 5.19 vs Con HATs * <i>p</i> = 0.040	135.2 ± 13.42
US, HAT (US HAT)	104.6 ± 14.43 vs US NT # <i>p</i> < 0.001	110.8 ± 8.03 vs US NT # <i>p</i> = 0.001	122.2 ± 12.60

Numbers of internalized GluA1, GluA2 and GluA3 puncta. Numbers of internalized GluA1-3 puncta in stressed non-treated mice and number of internalized GluA2 puncta in stressed mice treated with HAT-solvent were increased in comparison with control non-treated mice. HAT-treated stressed mice displayed a decrease of number of internalized GluA1-2 puncta in comparison to stressed non-treated group (**p* < 0.05 vs. non-treated controls, **p* < 0.05 vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used, data are expressed in percent from non-treated controls). US: ultrasound; HAT: herbal antioxidant treatment. Data are Mean ± SEM.

treatment on these measures ($F_{2,25} = 6.070$, $p = 0.007$; $F_{2,25} = 6.965$, $p = 0.004$; $F_{2,25} = 4.682$, $p = 0.027$, respectively). Number of internalized GluA1-A3 puncta were increased in stressed non-treated mice in comparison with control non-treated mice ($p < 0.001$ and $d = 2.93$; $p < 0.003$ and $d = 2.36$; $p = 0.043$ and $d = 1.67$, respectively, Tukey's and Cohen's d tests, Table 2). In stressed mice treated with HAT-solvent, in comparison with control non-treated group, number of internalized GluA2 puncta were significantly increased ($p = 0.040$ and $d = 2.65$), while number of internalized GluA1 and GluA3 puncta was not significantly different from non-stressed controls ($p = 0.057$; $p = 0.067$, respectively). HAT-treated stressed mice displayed a decrease of number of internalized GluA1 and GluA2, but not GluA3 puncta in comparison to stressed non-treated group ($p < 0.001$ and $d = 3.35$; $p = 0.001$ and $d = 2.78$; $p = 0.114$, respectively; Figure 3(A,C,E)). No changes in number of internalized GluA1 and GluA3 puncta were found in non-stressed mice treated with HAT-solvent, while above-presented data suggest similar to HAT effects of the solvent on AMPA receptor sub-units expression and internalization during stress.

ANOVA revealed no significant effect of stress and herbal treatment interaction on number of surface GluA1, GluA2 and GluA3 puncta ($F_{2,25} = 0.623$, $p = 0.545$; $F_{2,25} = 2.783$, $p = 0.082$; $F_{2,25} = 4.039$, $p = 0.08$, respectively). There was a significant effect of stress on number of surface GluA1-A3 puncta ($F_{1,25} = 18.45$, $p < 0.001$; $F_{1,25} = 6.649$, $p < 0.017$; $F_{1,25} = 5.928$, $p = 0.023$, respectively). There was a non-significant decrease in surface GluA1-A3 puncta in non-treated stressed mice ($p = 0.073$; $p = 0.190$; $p = 0.14$, respectively, Tukey's test; Figure 3(B,D,F) and Table 3) and in stressed mice treated with HAT-solvent ($p = 0.067$, $p = 0.168$; $p = 0.169$, respectively), in comparison with control non-treated mice. Herbal treatment significantly affected number of surface GluA2 and GluA3 puncta ($F_{2,25} = 4.98$, $p = 0.016$; $F_{2,25} = 5.992$,

$p = 0.008$, respectively), but not GluA1 puncta ($F_{2,25} = 1.229$, $p = 0.310$). Specifically, number of surface GluA2 and GluA3, but not GluA1 puncta were significantly increased in stressed HAT-treated mice in comparison with stressed non-treated mice ($p = 0.036$ and $d = 2.32$; $p = 0.007$ and $d = 2.78$, $p = 0.341$, respectively). No changes in these measures were found in HAT-treated non-stressed mice.

ANOVA revealed a significant interaction between the ultrasound stress and the herbal treatment in GluA1, GluA2, and GluA3 internalization index of subunits ($F_{2,24} = 12.19$, $p = 0.0002$; $F_{2,24} = 9.704$, $p = 0.0008$ and $F_{2,24} = 4.926$, $p = 0.0157$, respectively, two-way ANOVA). We also found a significant stress effect ($F_{1,24} = 72.02$, $p < 0.0001$; $F_{1,24} = 26.93$, $p < 0.0001$ and $F_{1,24} = 6.808$, $p = 0.0151$, respectively) and significant effect of the herbal treatment ($F_{2,24} = 3.784$, $p = 0.0367$; $F_{2,24} = 7.408$, $p = 0.003$ and $F_{2,24} = 6.808$, $p = 0.003$, respectively) on the internalization index. Post hoc analysis revealed a significant increase in the internalization index of GluA1, GluA2, and GluA3 in stressed non-treated mice in comparison with non-treated controls ($p < 0.0001$ and $d = 2.12$, $p = 0.003$ and $d = 1.14$, $p = 0.0419$ and $d = 1.02$, respectively, Tukey's and Cohen's d tests; Table 4). There was also non-significant decrease of the internalization index of GluA1 and GluA2, but not of GluA3 subunit in stressed mice treated with HAT-solvent in comparison with non-treated controls ($p = 0.0001$ and $d = 2.02$, $p = 0.0002$ and $d = 1.93$, $p = 0.157$ and $d = 1.13$, respectively). Internalization index was significantly decreased in stressed HAT-treated mice in comparison to stressed non-treated group for all subunits ($p = 0.0001$ and $d = 2.34$, $p < 0.0001$ and $d = 2.16$, $p = 0.012$ and $d = 1.09$, respectively). Thus, ultrasound exposure increased internalization of GluA1-A3 subunits of hippocampal AMPA receptor. Herbal treatment precluded these effects of the ultrasound stress in not-treated or HAT-solvent-treated animals and increased surface GluA2 and GluA3 subunits.

3.5. Ultrasound exposure and herbal treatment affect hippocampal expression of cytokines, GSK3 β , Akt and number of Ki67-positive cells

We found significant increases in pro-inflammatory cytokine expression. There was no significant effect of stress and treatment interaction on TNF mRNA expression ($F_{2,25} = 3.36$, $p = 0.051$, two-way ANOVA). ANOVA revealed a significant effect of stress ($F_{1,25} = 28.41$, $p < 0.001$) and herbal treatment ($F_{2,25} = 3.395$, $p = 0.049$) on this measure, which was increased in stressed non-treated mice ($p = 0.027$ and $d = 3.20$, Tukey's and Cohen's d test) and in stressed mice that received HAT-solvent in comparison with control non-treated mice ($p = 0.001$ and $d = 2.76$; Figure 4(A)).

We found a significant interaction between ultrasound exposure and herbal treatment on GSK-3 β and IL-1 β mRNA levels ($F_{2,25} = 3.999$, $p = 0.031$; $F_{2,25} = 5.860$, $p = 0.008$, respectively, two-way ANOVA). These measures were significantly altered by stress ($F_{1,25} = 45.87$, $p < 0.001$; $F_{1,25} = 32.48$, $p < 0.001$, respectively) and herbal treatment ($F_{2,25} = 6.104$, $p = 0.007$; $F_{2,25} = 4.088$, $p = 0.029$, respectively, Figure 4(B,C)). In comparison with control non-treated mice, GSK-3 β and IL-

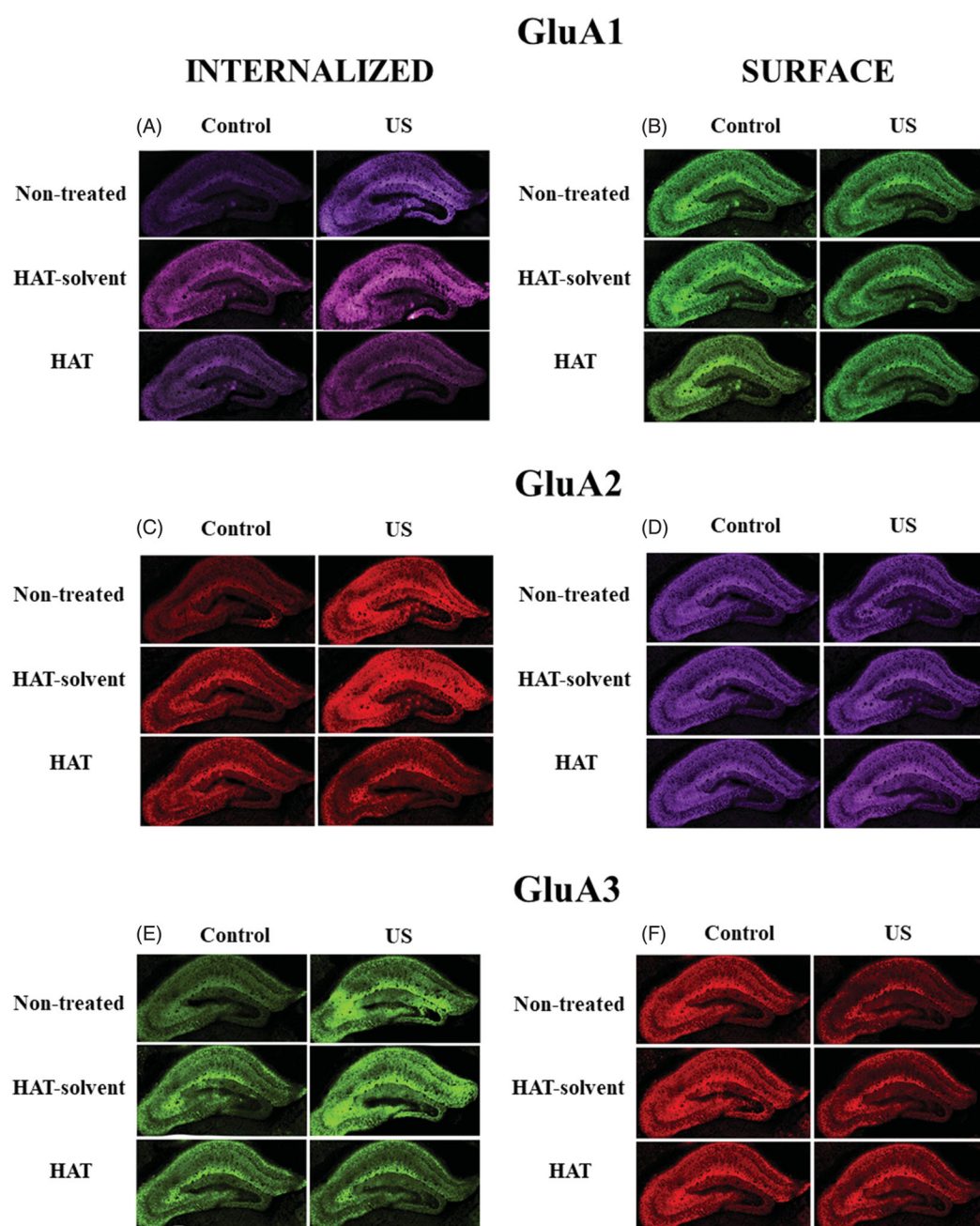


Figure 3. Immunohistochemical analysis of surface and internalized GluA subunits. Immunohistochemical analysis of (A) internalized GluA1, (C) GluA2 and (E) GluA3 subunits and surface (B) GluA1, (D) GluA2 and (F) GluA3 subunits of AMPA receptor.

1 β mRNA levels were increased in stressed non-treated mice ($p < 0.001$ and $d = 2.75$; $p < 0.001$ and $d = 3.45$, respectively, Tukey's and Cohen's d tests) and stressed mice that received HAT-solvent ($p = 0.002$ and $d = 2.45$; $p = 0.002$ and $d = 2.16$, respectively). There was a significant decrease in these parameters in stressed HAT-treated mice in comparison with stressed non-treated mice ($p = 0.002$ and $d = 2.48$; $p = 0.006$ and $d = 3.31$, respectively; Figure 4(B,C)).

ANOVA revealed no significant stress and treatment interaction on mRNA levels IL-6, IL-15 and Akt ($F_{2,25} = 2.950$, $p = 0.071$; $F_{2,25} = 0.02$, $p = 0.979$; $F_{2,25} = 0.433$, $p = 0.653$, respectively, two-way ANOVA) in the hippocampus. Stress significantly increased mRNA concentrations of IL-6 and Akt, but

not IL-15 mRNA levels in the hippocampus ($F_{1,25} = 35.66$, $p < 0.001$; $F_{1,25} = 6.250$, $p = 0.019$; $F_{1,25} = 0.593$, $p = 0.449$, respectively). Furthermore, ANOVA revealed no significant effect of herbal treatment on these measures ($F_{2,25} = 1.357$, $p = 0.276$; $F_{2,25} = 0.1791$, $p = 0.837$; $F_{2,25} = 0.846$, $p = 0.441$, respectively). Stressed non-treated group displayed significantly increased IL-6 mRNA levels in comparison with control non-treated mice ($p < 0.001$ and $d = 1.32$, Tukey's and Cohen's d tests, Figure 4(D)). At the same time, post hoc analysis did not reveal significant differences in HAT-treated stressed mice in this measure ($p = 0.363$; $p = 0.567$, Tukey's test, respectively). Stressed mice treated with HAT-solvent showed no changes in IL-6 mRNA level in comparison with control non-treated

Table 3. Numbers of surface GluA1, GluA2 and GluA3 puncta.

Groups	Surface puncta, % from control		
	GluA1	GluA2	GluA3
Control, non-treated (Con NT)	100 ± 15.97	100 ± 14.06	100 ± 15.26
Control, HAT-solvent (Con HATs)	113.6 ± 7.29	97.6 ± 9.65	99.2 ± 7.93
Control, HAT (Con HAT)	108.6 ± 10.47	106 ± 11.46	90 ± 16.02
US, non-treated (US NT)	46 ± 6.47	65 ± 18.9	61.4 ± 4.76
US, HAT-solvent (US HATs)	58.2 ± 9.59	61.2 ± 7.69	53.6 ± 7.95
US, HAT (US HAT)	79.2 ± 13.85	112.4 ± 9.82 vs US NT # <i>p</i> = 0.016	120.6 ± 14.6 vs US NT # <i>p</i> = 0.008

Numbers of surface GluA1, GluA2 and GluA3 puncta. Number of surface GluA2-3 puncta were significantly increased in stressed HAT-treated mice in comparison with stressed non-treated mice (**p* < 0.05 vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used, data are expressed in percent from non-treated controls). US: ultrasound; HAT: herbal antioxidant treatment. Data are Mean ± SEM.

animals (*p* = 0.978, *p* = 0.786; **Figure 4(D)**) suggesting partial anti-inflammatory action of the solvent, which might underlie its effects on behavioral parameters.

There was no significant interaction between stress in HAT treatment in their effects on the number of Ki67-positive cells, a marker of neuronal proliferation ($F_{2,24} = 0.607$, *p* = 0.553, two-way ANOVA). This parameter was significantly affected by stress ($F_{1,24} = 30.38$, *p* < 0.001) and herbal treatment ($F_{2,24} = 6.603$, *p* = 0.005). Post hoc analysis revealed a decrease in the number of Ki67-positive cells in stressed non-treated mice (*p* = 0.017 and *d* = 3.18, Tukey's and Cohen's *d* tests) and in stressed mice that received HAT-solvent in comparison with control non-treated mice (*p* = 0.013 and *d* = 2.79, **Figure 4(G)**). We found significant effect of stress ($F_{1,24} = 9.701$, *p* = 0.005), but not treatment ($F_{2,24} = 0.159$, *p* = 0.854), nor interaction ($F_{2,24} = 0.939$, *p* = 0.405, two-way ANOVA, **Supplementary Figure 6**) on the number of NeuN-positive cells. Post hoc analysis did not reveal significant changes in stressed non-treated mice and stressed mice treated with HAT-solvent in comparison with control (*p* = 0.2247; *p* = 0.1978, respectively, Tukey's test). Thus, ultrasound exposure resulted in the decreased number of Ki67-positive cells, but did not affect the number of NeuN-positive cells.

3.6. Elevated hippocampal malondialdehyde content correlates with neurobiological effects of ultrasound stress: effects of herbal treatment

We found a significant interaction between stress and treatment in the effects on MDA content ($F_{2,25} = 4.360$, *p* = 0.024). There was a significant effect of stress ($F_{1,25} = 23.16$, *p* < 0.001) and herbal treatment ($F_{2,25} = 4.556$, *p* = 0.02) on the MDA levels in the hippocampus. Post hoc analysis revealed elevated hippocampal MDA content in stressed non-treated mice (*p* = 0.033 and *d* = 1.90, Tukey's and Cohen's *d* tests) and stressed mice that received HAT-solvent in comparison with control non-treated group (*p* = 0.002 and *d* = 2.16). There was a significant decrease in this parameter in stressed HAT-treated mice in comparison with stressed non-treated mice (*p* = 0.034 and *d* = 2.09, **Figure 5(A)**).

Table 4. Internalization index of GluA1, GluA2 and GluA3 subunits.

Groups	Internalization index, %		
	GluA1	GluA2	GluA3
Control, non-treated (Con NT)	49.2 ± 6.1	49.5 ± 5.18	49.6 ± 6.83
Control, HAT-solvent (Con HATs)	44.9 ± 2.19	50.4 ± 2.69	51.1 ± 1.17
Control, HAT (Con HAT)	51.9 ± 7.47	45.8 ± 3.97	46.4 ± 4.46
US, non-treated (US NT)	82.9 ± 2.71	75.1 ± 4.06	72.1 ± 3.36
vs Con NT	* <i>p</i> < 0.0001	* <i>p</i> = 0.003	* <i>p</i> = 0.0419
US, HAT-solvent (US HATs)	73.8 ± 4.17	73.2 ± 2.09	71.1 ± 4.88
vs Con NT	* <i>p</i> < 0.0001	* <i>p</i> = 0.0002	* <i>p</i> = 0.157
US, HAT (US HAT)	56.7 ± 7.19	49.7 ± 3.73	50.4 ± 4.61
vs US NT	* <i>p</i> = 0.0001	* <i>p</i> < 0.0001	* <i>p</i> = 0.012

Internalization index of GluA1, GluA2 and GluA3 subunits. Internalization index of GluA1-3 subunits in stressed non-treated mice and stressed mice treated with HAT-solvent was significantly increased in comparison with non-treated controls. This parameter was significantly decreased in stressed HAT-treated mice in comparison to stressed non-treated group in all subunits (**p* < 0.05 vs. non-treated controls, #*p* < 0.05 vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used). US: ultrasound; HAT: herbal antioxidant treatment. Data are Mean ± SEM.

Correlation analysis performed in all experimental groups reveal significant correlations between hippocampal MDA levels and the most of investigated end-points. Significant correlations were found between the MDA content and the total number of entries to the open arms of the O-maze (*p* = 0.002, *r* = −0.532), the time spent therein (*p* < 0.001, *r* = −0.573), the total number of rearings (*p* < 0.001, *r* = −0.632), the time spent in the center of the open field (*p* < 0.001, *r* = −0.635), latency to attack (*p* < 0.001, *r* = −0.734), the total number of attacks (*p* = 0.013, *r* = 0.551; **Figure 5(B–G)**). There were significant correlations between the MDA content and mRNA levels of GluA2 (*p* = 0.001, *r* = 0.551), TNF (*p* < 0.001, *r* = 0.807), IL-1β (*p* < 0.001, *r* = 0.649), and GSK3β (*p* = 0.002, *r* = 0.543; **Figure 5(H–K)**). Finally, we found significant correlations between the MDA content and internalized GluA1 puncta (*p* < 0.001, *r* = 0.639), internalized GluA2 puncta (*p* = 0.002, *r* = 0.569) and number of Ki67-positive cells (*p* < 0.001, *r* = −0.56; **Figure 5(L–N)**). Isolated analysis of the above-described correlations showed significant correlations of MDA content with most of the indicated parameters in stressed, but not in non-stressed groups analyzed (see **Supplementary material**, **Supplementary Figures 7 and 8**). This further highlights a relationship between ultrasound stress-induced neurobiological changes and oxidative stress.

Discussion

Exposure to ultrasound stress resulted in elevated scores of correlates of anxiety-like behavior in the non-treated group of mice. These changes were accompanied by suppressed vertical/exploratory activity, augmented aggression scores, grooming behavior, hippocampal expression of the AMPA receptor subunit GluA2, and of internalization rate of GluA1-A3 subunits of this receptor. We observed an increase in hippocampal expression of the investigated pro-inflammatory cytokines IL-1β, IL-6, TNF, as well as of GSK3β, and elevated malondialdehyde content. The number of Ki67-positive cells,

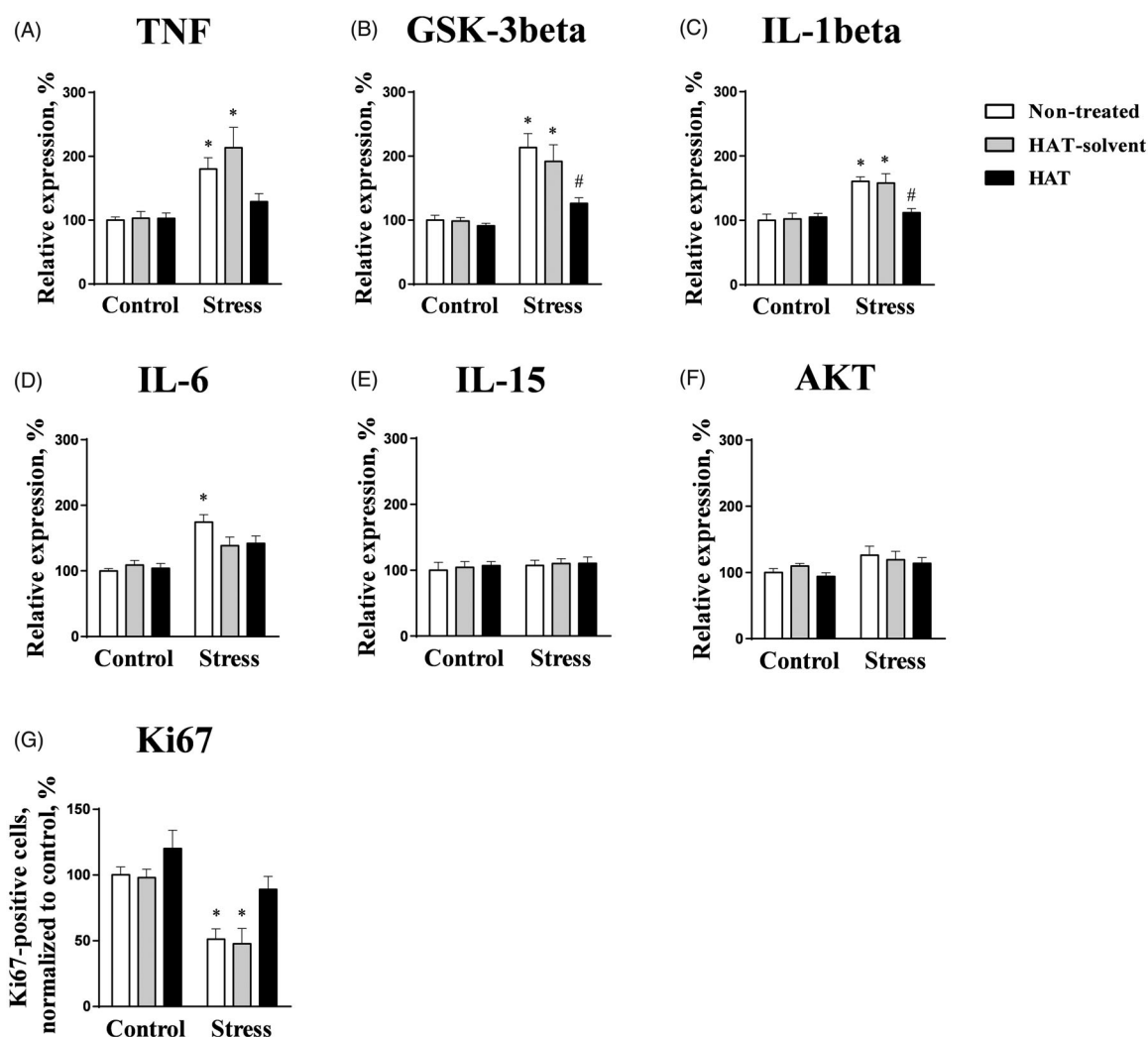


Figure 4. Ultrasound exposure and herbal treatment effect on expression of cytokines, GSK3 β , Akt and number of Ki67-positive cells in the hippocampus. Stressed non-treated mice and stressed mice treated with HAT-solvent displayed a significant increase of (A) TNF mRNA expression, (B) GSK-3 β mRNA expression, (C) IL-1 β mRNA expression and (D) IL-6 mRNA expression in the hippocampus, in comparison with control non-treated mice. (E) There were no significant differences between groups in IL-15 mRNA (F) and in Akt mRNA concentrations. (G) Stressed non-treated mice and stressed mice treated with HAT-solvent displayed a significant decrease in Ki67-positive cells in the hippocampus that was not observed in HAT-treated group subjected to stress. * $p < 0.05$ vs. non-treated controls, # $p < 0.05$ vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test. Six to eight animals per group were used. HAT: herbal antioxidant treatment. Bars are Mean $\pm$ SEM.

a marker of cellular proliferation, was reduced in stressed non-treated animals. Chronic dosing with herbal antioxidant/anti-inflammatory treatment prevented most of these changes. This study revealed a strong correlation between the hippocampal malondialdehyde levels on one hand, and behavioral and molecular changes induced by ultrasound stress on another hand. These correlations were only present in stressed, but not in non-stressed mice which further highlights the role of oxidative stress in neurobiological abnormalities, including behavioral correlates of elevated anxiety associated with a state of "emotional" stress.

Behavioral parameters of anxiety-like state in the O-maze were elevated in non-treated animals exposed to ultrasound. This finding was further supported by the results obtained in the open field test and is in line with previous findings in this paradigm (Pavlov et al., 2019). Increase in grooming behavior and decrease in vertical activity in stressed non-treated mice observed in this study can be a sign of chronic stress in small rodents and is associated with behavioral stereotypy and

suppressed novelty exploration, respectively (Kalueff, Keisala, Minasyan, Kuuslahti, & Tuohimaa, 2006; Strekalova, Spanagel, Bartsch, Henn, & Gass, 2004). Augmented aggression scores observed in the present study are consistent with previous results obtained in the ultrasound stress model (Gorlova et al., 2019; Pavlov et al., 2019) and are generally supported by the literature demonstrating a relationship between increases in behavioral parameters of stress, anxiety and aggression both in animal and clinical studies (Bandelow & Michaelis, 2015; Chen et al., 2013). A lack of changes in general locomotion rules out a possibility of confounds that potentially can result from prolonged exposure to stress (Strekalova et al., 2005), in behavioral analysis of mice.

Emotional-like abnormalities induced by ultrasound in mice were accompanied by altered gene expression of the AMPA receptor subunits, suggesting their dysregulation, in the hippocampus. A dysregulation of these processes in the brain was previously shown to be related with increased signs of anxiety (Weber et al., 2015), aggression (Adamczyk

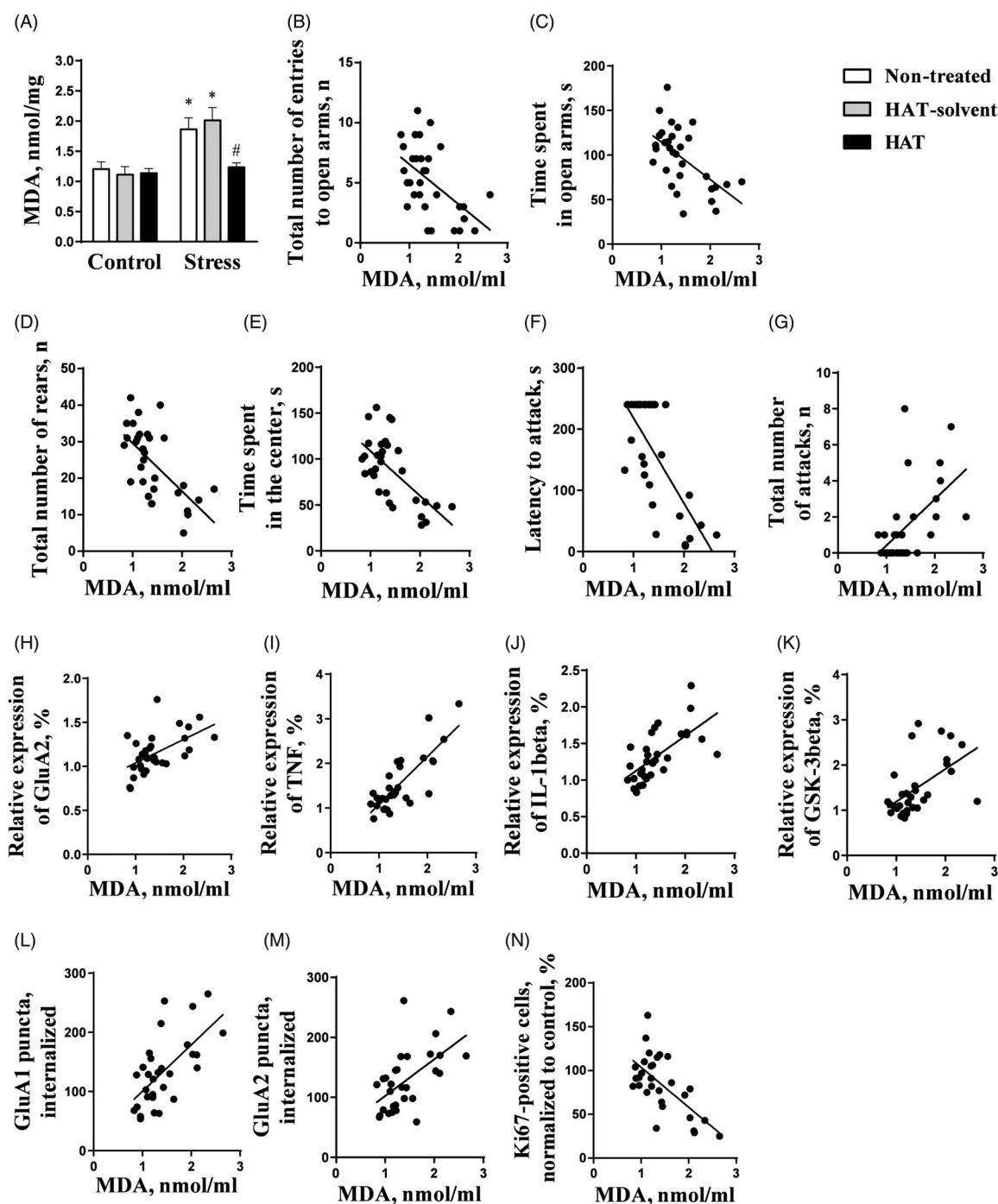


Figure 5. Elevated hippocampal malondialdehyde content correlates with neurobiological effects of ultrasound stress: effects of herbal treatment. (A) Stressed non-treated mice and stressed mice treated with HAT-solvent displayed a significant increase in MDA content in the hippocampus. Stressed HAT-treated animals did not differ from non-treated mice in this parameter and displayed a significantly decreased MDA levels in comparison with stressed non-treated group. Significant correlations were found between the malondialdehyde content and (B) total number of entries to open arms of the O-maze, (C) time spent in open arms of the O-maze, (D) total number of rears, (E) time spent in the center of the open field, (F) latency to attack, (G) total number of attacks, (H) GluA2 mRNA expression, (I) TNF mRNA expression, (J) IL-1 β mRNA expression, (K) GSK-3 β mRNA expression, (L) GluA1 internalized puncta, (M) GluA2 internalized puncta, (N) number of Ki67-positive cells. * $p < 0.05$ vs. non-treated controls, # $p < 0.05$ vs. non-treated stressed group, two-way ANOVA and post hoc Tukey's test; Spearman correlation. Six to eight animals per group were used. HAT: herbal antioxidant treatment. Bars are Mean $\pm$ SEM.

et al., 2012) and impaired recovery from stress (Martinowich et al., 2012). We found an increased gene expression of GluA2 subunit and a non-significant decrease of GluA1 and GluA3 subunits expression, which is in line with previous studies (Gorlova et al., 2019). Other stress experiments, e.g. with mouse unpredictable chronic stress paradigm, revealed up-regulated GluA2 expression in the brain associated with

signs of increased anxiety-like and aggressive behaviors (Costa-Nunes et al., 2014).

Immunohistochemical analysis of expression of internalized and surface GluA1-3 subunits uncovered an elevated rate of internalization of subunits of all investigated AMPA receptor subunits in the hippocampus in stressed non-treated mice. The internalization of the AMPA receptor subunits plays

a key role in the modulation of receptor bioavailability, and thus, functioning, as this process terminates receptor cell surface signaling and designates activated receptors for degradation (Pradhan, Tawfik, Laboy, & Scherrer, 2016; van Rijnsoever, Sidler, & Fritschy, 2005). The internalization regulates cell-surface expression of the AMPA receptor in neurons (van Rijnsoever et al., 2005). In the present study, increased internalization and the increased ratio of internalized/surface AMPA receptor subunits may suggest an adaptive decrease in the glutamate-mediated neurotransmission via GluA1-3 subunits. This, in turn, can dampen the negative effects of glutamatergic excitotoxicity, an important pathogenetic factor of stress (Caudal, Godsil, Mailliet, Bergerot, & Jay, 2010; Bonini et al., 2016). The reported increase in the mRNA levels of GluA2 may be a sign of compensatory changes triggered by downregulation of receptor function. In this context, lack of similar increases in gene expression of GluA1 and GluA3 subunits might be explained by different dynamics of these regulatory processes.

Thus, present findings suggest a suppressive effect of ultrasound stress on the AMPA receptor-mediated signaling and synaptic plasticity that can underlie stress-related emotional abnormalities (Bowers, Chen, & Bonci, 2010) and explain reported here behavioral changes. Chronic stress can impair adult neurogenesis (Schoenfeld & Gould, 2012; Lajud, Roque, Cajero, Gutiérrez-Ospina, & Torner, 2012; Vignisse et al., 2017), and reduced number of Ki67-positive cells in the hippocampus serves as further evidence for suppressed neuronal plasticity induced by ultrasound exposure. This finding is in line with previously reported decreases in gene and protein expression of plasticity regulators PSD95, PSA-NCAM, β -catenin (Gorlova et al., 2019) and density of Ki67-positive cells in the subgranular zone of the dentate gyrus of the hippocampus (Pavlov et al., 2019).

We found an increased hippocampal expression of pro-inflammatory cytokines IL-1 β , IL-6 and TNF, that supports our previous observations. On another hand, the expression of the anti-inflammatory cytokine IL-15 was unaltered. As neuro-inflammation was shown to accompany pathological anxiety (Rossi et al., 2017; Salim, 2017) and since an increase in the synthesis of brain pro-inflammatory cytokines in mice can promote anxiogenic-like (Chen et al., 2013) and aggressive behaviors (Takahashi, Flanigan, McEwen, & Russo, 2018), present data implicate the reported here over-production of pro-inflammatory cytokines in the anxiogenic effects of ultrasound stress. We also demonstrated that in the ultrasound-exposed mice, elevated scores of anxiety-like behaviors were associated with the over-expression of GSK-3 β , that governs the production of pro-inflammatory cytokines (Green & Nolan, 2012), affects oxidative stress (Potz et al., 2018) and regulates hippocampal cellular plasticity (Peineau et al., 2009). In current study, the expression of its physiological antagonist Akt (Manning & Toker, 2017), was unaltered.

The use of hippocampi from the same animals for the variety of assays employed here made it possible to perform the correlation analysis between the pro-oxidative changes and other molecular end-points in addition to behavioral parameters. Most of the investigated neurobiological parameters correlated with hippocampal malondialdehyde content. There were

significant correlations between this measure and scores of anxiety-like behaviors, as well as open field exploratory and stereotypic behaviors, and parameters of aggression. Malondialdehyde levels significantly correlated with the expression of GluA2 subunit and internalization of GluA1 and GluA2 subunits of AMPA receptor, gene expression of IL-1 β , TNF, GSK3 β , and the number of Ki67-positive cells. Besides, positive correlations were found between the number of attacks and most of investigated markers of cellular plasticity and neuroinflammation, as well as MDA concentration in the hippocampus. They are likely to be due to the independent effects of the ultrasound stress on both aggression and stress-related markers, rather than the effect of a single experience of aggressive interaction alone. Recent studies revealed a decreased number of Ki67-positive cells in the hippocampus of a mouse strain that displays a reduction of aggressive behavior after ultrasound exposure (Gorlova, *personal communication*). Notably, analysis of non-stressed groups alone did not reveal any significant correlations between MDA levels and above-mentioned parameters (Supplementary Figure 7). These results further suggest the importance of pro-oxidative and pro-inflammatory changes as mechanisms of the negative effects of ultrasound stress in mice, and the role of the hippocampus in these changes.

Chronic treatment of mice with applied herbal composition precluded ultrasound-induced increases of manifestations of anxiety-like and aggressive behaviors, as well as most of the reported molecular changes pointing to the anti-oxidant and anti-inflammatory effects of the composition. Specifically, administration of herbal composition has prevented the increase in malondialdehyde content and over-expression of pro-inflammatory cytokines in ultrasound-exposed mice. This was accompanied by normalization of evaluated behavioral correlates of emotionality-like states, AMPA receptor subunits and expression of a marker of distress GSK3 β and plasticity, Ki67, in the hippocampus. Application of herbal composition in non-stressed mice and the use of its solvent in stressed animals did not alter the majority of investigated parameters.

Previous studies demonstrated ameliorative effects of thiamin compounds with anti-oxidant and anti-inflammatory properties on the parameters of emotionality-like states and plasticity during stress (Markova et al., 2017; Vignisse et al., 2017; Pavlov et al., 2019). Other studies revealed anxiolytic-like effects of compounds with antioxidant properties (Bystritsky et al., 2012; Dourado & Cesar, 2015). In addition, a relationship between the anti-anxiety action and normalization of malondialdehyde content in the brain was demonstrated in mice (Sulakhiya et al., 2016). Thus, the beneficial functional effects of the herbal composition used in this study can be explained by its antioxidant and anti-inflammatory properties.

Treatments with the components of the herbal composition used in this study were previously shown to exert similar neurobiological effects: e.g. polysaccharides from red seaweed suppressed expression of TNF receptor-associated factor-6, protecting against LPS-induced toxicity (Cui et al., 2019); domestic brown seaweeds reduced expression of pro-inflammatory cytokines in activated macrophages (Kwon et al., 2013); ginger phenolics diminished lipid peroxidation and oxidative stress

neuronal damage in rats (Vipin, Rao, Kurrey, Anu, & Venkateswaran, 2017); vanillin suppressed free radical production in a mouse model of cancer (Tai, Sawano, Yazama, & Ito, 2011). Flavonoids from citrus fruits were shown to enhance activity of anti-oxidant defense enzymes, such as superoxide dismutase, catalase and glutathione (Zaidun, Thent, & Latiff, 2018), prevent stress-induced neurotoxicity, oxidative stress (Heo et al., 2008), and over-production of inflammatory factors, including C-reactive protein, TNF, and IL-1 β (Dourado & Cesar, 2015). Elderberry extract normalized the activities of superoxide dismutase and glutathione peroxidase (Salvador et al., 2016), and reduced serum IL-6 levels (Youdim, Martin, & Joseph, 2000). Finally, glutamine, a precursor of glutathione, was shown to regulate cellular redox balance (Matés, Pérez-Gómez, Núñez de Castro, Asenjo, & Márquez, 2002) and prevent changes in gene expression associated with oxidative stress in rats (Tsai, Liu, Yeh, Chiu, & Yeh, 2012).

The administration of its solvent resulted in partial amelioration of several measures of behavior, IL-6 and GluA2 expression and GluA3 internalization that was reminiscent of the effects of HAT itself and is likely due to well-known anti-stress effects of alcohol (Wang et al., 2008), its main component. As chronic intake of small doses of alcohol can induce anti-inflammatory effects (Stote, Tracy, Taylor, & Baer, 2016; Mangnus et al., 2018) and alters AMPA receptor-mediated signaling in rodents (Ary et al., 2012; Salling et al., 2016), HAT-solvent might produce additive effects in the action of investigated here herbal cocktail.

Conclusion

Altogether, presented data emphasize the role of oxidative stress and neuroinflammation in neurobiological abnormalities associated with “emotional” stress-like state and highlight the beneficial anti-anxiety properties of the applied herbal composition. Generally, pathological changes in oxidative stress, pro-inflammatory, glutamatergic, and plasticity mechanisms during stress are inter-related, and metabolic/mitochondrial regulations can be viewed as pivotal in these dysregulatory cascades (Hussain et al., 2016; Picard & McEwen, 2018; Rezin et al., 2008). Given that patients with anxiety disorders revealed a reduction in brain markers of plasticity, signs of neuroinflammation (Michopoulos, Powers, Gillespie, Ressler, & Jovanovic, 2017), and aberrant AMPA-receptor mediated neurotransmission (Bergink, van Megen, & Westenberg, 2004), it can be anticipated that the investigated herbal composition has a therapeutic potential not only for prevention of stress-related anxiety, but also under clinical conditions of elevated anxiety.

While the number of elements in the herbal composition is a limitation of this study, as the question whether any individual element or combinations of elements contribute to the effects could not be addressed, present work highlights several mechanisms related to glutamatergic neurotransmission, inflammation and oxidative stress, which are implicated in the anti-stress action of tested here herbal cocktail. Further studies are required to elucidate these mechanisms more in a detail.

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